Extreme learning machine-assisted solution of biharmonic equations via its coupled schemes

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Abstract

The application of machine learning techniques to solve partial differential equations has garnered increasing attention in the fields of scientific computation and engineering applications. This study introduces a novel approach, the Coupled Extreme Learning Machine (CELM), which incorporates physical laws to address a specific class of fourth-order biharmonic equations. The reformulation transforms the problem into two well-posed Poisson problems. Various activation functions, such as tangent, Gaussian, sine, and trigonometric (sin+cos) functions, are employed to evaluate the performance of the CELM method. Notably, the sine and trigonometric functions exhibit a remarkable ability to minimize the approximation error of the CELM model effectively. The study also explores different initialization approaches for the weights and biases of the hidden units in the CELM model and determines the optimal number of hidden units through numerical experiments. The results demonstrate that the proposed CELM algorithm is exact and efficient in addressing biharmonic equations in both regular and irregular domains.

AMS subject classifications 35J58 · 65N35 · 68T07

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1. Introduction

Partial differential equations (PDEs), which have their roots in science and engineering, have been instrumental in advancing the fields of physics, chemistry, biology, image processing, and more. Analytical solutions for PDEs are often elusive, necessitating the reliance on numerical methods and approximation techniques for their resolution. This paper focuses on the numerical solution of the following type of biharmonic equation

$$\begin{cases} \Delta^2 u(\boldsymbol{x}) = f(\boldsymbol{x}), \ \text{in } \Omega, \\ u(\boldsymbol{x}) = g(\boldsymbol{x}), \ \text{on } \partial\Omega, \\ \Delta u(\boldsymbol{x}) = h(\boldsymbol{x}), \ \text{on } \partial\Omega. \end{cases} \tag{1}$$

where Ω is a polygonal or polyhedral domain $d-$dimensional in Euclidean space with piecewise Lipschitz boundary which satisfies an interior cone condition, $f(\boldsymbol{x}) \in L^2(\Omega)$ is a prescribed function and Δ stands for a standard Laplace operator. The expressions for the operators $\Delta u(\boldsymbol{x})$ and $\Delta^2 u(\boldsymbol{x})$ are given by

$$\Delta u(\boldsymbol{x}) = \sum_{i=1}^{d} \frac{\partial^2 u}{\partial x_i^2}$$

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and

$$\Delta^2 u(\boldsymbol{x}) = \sum_{i=1}^{d} \frac{\partial^4 u}{\partial x_i^4} + \sum_{i=1}^{d}\sum_{j=1}^{d} \frac{\partial^4 u}{\partial x_i^2 x_j^2},$$

respectively.

The biharmonic equation, a high-order partial differential equation, is frequently used in physics and applied mathematics, especially in the study of elasticity and Stokes flow problems. Its applications range from scattered data fitting with thin plate splines [1] to fluid mechanics [2, 3] and linear elasticity [4, 5]. Over recent decades, various traditional numerical methods have been developed to resolve this equation (1), and these methods can be broadly classified into two categories: the direct (uncoupled) method and the coupled (splitting, mixed) method.

Concerning the direct approach, diverse methodologies are applied to address the biharmonic equation (1). These methods encompass finite difference approaches embracing the uncoupled scheme [6–9], finite volume techniques [10–12], and finite element methods (FEM) basing on the variational formulation of biharmonic problem such as non-conforming FEM [13–15] and conforming FEM [16, 17]. The key idea of coupled approach is decomposing the biharmonic equation into two interconnected Poisson equations by introducing some auxiliary variables. Through this coupled scheme, diverse strategies can be employed to solve the two second-order equations, including finite difference methods [9, 18, 19], finite element methods, and mixed element methods [20–24]. Additionally, the collocation method [25–27] and the radial basis functions (RBF) method [28–30] represent potential approaches for solving the biharmonic equation (1).

However, conventional methods may face challenges when dealing with irregular domains and high dimensions, often referred to as the curse of dimensionality. Machine learning, particularly artificial neural networks (ANNs), has emerged as a promising alternative for numerically approximating the solutions of differential equations, circumventing the limitations of traditional numerical techniques. Various types of ANNs, including extreme learning machines (ELM) and deep neural networks (DNN), have been developed to address PDEs, including the biharmonic equation (1). Several physics-informed neural network approaches have been proposed, directly applying the original formulation of (1). These approaches minimize loss and incorporate the governed residual term and corresponding boundaries [31–33]. Energy-based methods, developed according to the energy functional and embracing the variational formulation of (1) or its coupled scheme with a deep Ritz method [34–37], have also been explored. However, these DNN-based methods have exhibited drawbacks such as limited accuracy, unknown convergence, and slow training speed. Recently, a physics-informed Extreme Learning Machine (PIELM) method was introduced, incorporating physical laws and ELM for solving various PDEs [38–41]. PIELM extends the advantages of ELM, offering mesh-free characteristics, faster learning speed, limited increase in computational complexity with a growing number of collections, and compatibility with parallel architecture. ELM itself represents a special case of Randomized Neural Networks (RNN [42, 43]), where parameters are randomly assigned to links between hidden layers and then fixed during training. Integrating the RNN model with physical laws has led to the development of physics-based RNN models designed to solve various types of PDEs [44, 45].

In this work, we deal with the fourth-order elliptic equation (1) through a combination of the ELM method and the coupled scheme for the biharmonic equation, leading to the establishment of a coupled extreme learning machine (referred to as CELM) architecture. We further examine the influence of the activation function—comprising the tangent, Gaussian, sine, and trigonometric functions—on the performance of the CELM model. Additionally, we explore various initialization approaches for the parameters of hidden units, including Xavier initialization, standard normal initialization, and uniform initialization. The numerical results indicate that sine and trigonometric functions enhance the model's performance and exhibit robust capabilities.

The paper is organized as follows. In Section 2, we provide a brief introduction to the fundamental concept of an extreme learning machine. Section 3 details the construction of the CELM algorithm for approximating the solution of the biharmonic equation, incorporating its coupled scheme, activation function selection, and the initialization of internal parameters for hidden units. In Section 4, we present several numerical examples to evaluate the performance of the proposed CELM model, focusing on aspects such as accuracy and efficiency. Finally, concise conclusions are drawn in Section 5.

2. Extreme learning machine and its mathematical expression

This section offers a concise overview of the pertinent concepts and mathematical formulation of the Extreme Learning Machine (ELM). ELM belongs to the category of single hidden layer fully connected neural networks (SHFNN) within artificial neural networks, designed to define a mapping: $\mathbb{R}^d \to \mathbb{R}$. The distinguishing feature of ELM lies in its random assignment of input layer weights and biases, with the output weights analytically determined [46, 47]. Mathematically, the ELM mapping $\ell(\boldsymbol{x})$ with an input $\boldsymbol{x} \in \mathbb{R}^d$ is in the form [48] of

$$\ell(\boldsymbol{x}) = \sum_{i=1}^{m} \beta_i \sigma_i(\boldsymbol{W}^{[i]}\boldsymbol{x} + b_i), \tag{2}$$

where $\boldsymbol{W} = \left(\boldsymbol{W}^{[1]}, \boldsymbol{W}^{[2]}, \cdots, \boldsymbol{W}^{[m]}\right)^T \in \mathbb{R}^{m\times d}$ with $\boldsymbol{W}^{[i]} \in \mathbb{R}^d$ being the weight of the i_{th} hidden unit for ELM and $\boldsymbol{b} = (b_1, b_2, \cdots, b_m)^T \in \mathbb{R}^m$ is its corresponding bias vector. $\sigma(\cdot)$ is an element-wise non-linear function, commonly known as the activation function. $\boldsymbol{\beta} = [\beta_1, \beta_2, \cdots, \beta_m]^T \in \mathbb{R}^m$ is the weight vector connecting from the hidden units of ELM to its output. A schematic diagram of ELM with m hidden units is depicted in Figure 1.

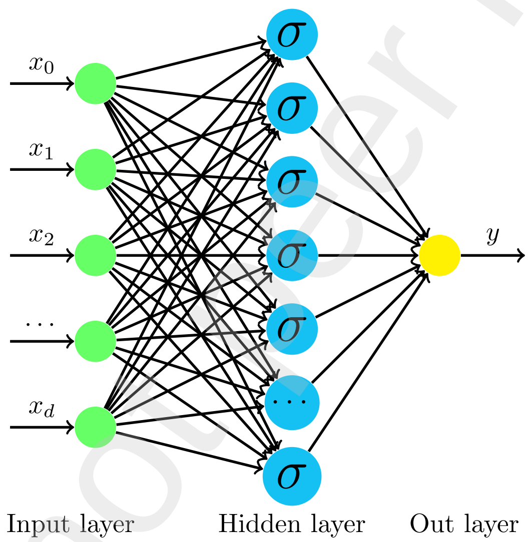


Figure 1: Basic structure of ELM.

When the ELM model incorporates physical laws, a novel approach is introduced, termed the Physically Informed Extreme Learning Machine (PIELM). PIELM optimizes its associated losses, which include information about the governed term and enforced boundary or initial conditions, by analytically calculating the outer layer weights. Unlike Physics-Informed Neural Network (PINN) methods employed for solving PDEs, PIELM effectively addresses several drawbacks inherent in PINNs, such as limited accuracy, unknown convergence, and a substantial computational burden leading to slow training speeds. Similar to previous ELMs utilized in classifications and fitting tasks, PIELM remains data-efficient and exhibits a faster performance compared to iterative gradient descent-based methods. Furthermore, PIELM retains the key properties of ELM, acting as a universal approximator for any continuous function; Huang et al. [48] rigorously proved that for any infinitely differentiable activation function SLFNs with N hidden nodes can learn N distinct samples exactly. This also implied that if a learning error (ϵ) is allowed, SLFNs will only require less than N hidden nodes.

3. Unified coupled ELM architecture to biharmonic equations

3.1. CELM algorithm to solve biharmonic equations

This section provides a detailed introduction to the CELM method for solving the biharmonic equation (1). Initially, the original fourth-order equation is transformed into two Poisson problems by introducing an auxiliary variable $v = \Delta u$. This rewriting is achieved using decoupled techniques commonly employed in traditional numerical methods like the Finite Element Method (FEM) and Finite Difference Method (FDM). The two Poisson equations are expressed as follows:

$$\begin{cases} \Delta v = f, & \text{in} \quad \Omega \\ \quad v = h, & \text{on} \quad \partial\Omega \end{cases} \quad \text{and} \quad \begin{cases} \Delta u = v, & \text{in} \quad \Omega \\ \quad u = g, & \text{on} \quad \partial\Omega \end{cases} \tag{3}$$

respectively. Instead of seeking an approximation for the unique solution of the original problem (1), our goal is to find a pair of approximations (u, v) that meet specific conditions or constraints. This approach is intended to minimize computational resources and time complexity. To solve the above couple PDEs, two ansatzes expressed by ELM are designed to approximate their solutions, they are:

$$\boldsymbol{u}(\boldsymbol{x}; \widetilde{\boldsymbol{W}}, \widetilde{\mathbf{b}}) = \sigma(\boldsymbol{x}^T \cdot \widetilde{\boldsymbol{W}} + \widetilde{\boldsymbol{b}}^T) \cdot \widetilde{\boldsymbol{\beta}}$$

and

$$\boldsymbol{v}(\boldsymbol{x}; \widehat{\boldsymbol{W}}, \widehat{\mathbf{b}}) = \sigma(\boldsymbol{x}^T \cdot \widehat{\boldsymbol{W}} + \widehat{\boldsymbol{b}}^T) \cdot \widehat{\boldsymbol{\beta}},$$

respectively. Substituting $\boldsymbol{v}(\boldsymbol{x}; \widehat{\boldsymbol{W}}, \widehat{\mathbf{b}})$ into the former equation of (3) for v, we can obtain the following equation

$$\begin{cases} \Delta \left(\sum_{i=1}^{K} \widehat{\beta}_i \sigma\big(V_i(\boldsymbol{x})\big) \right) = f(\boldsymbol{x}) \\ \sum_{i=1}^{K} \widehat{\beta}_i \sigma\big(V_i(\boldsymbol{x})\big) = h(\boldsymbol{x}) \end{cases} \tag{4}$$

with $V_i(\boldsymbol{x}) = \boldsymbol{x}^T \cdot \widehat{\boldsymbol{W}}^{[i]} + \widehat{b}_i$ being the linear output for i_{th} hidden unit of ELM and K is the number of hidden unit. Since the latter equation of (3) is related to u and $\boldsymbol{v}(\boldsymbol{x}; \widehat{\boldsymbol{W}}, \widehat{\mathbf{b}})$ is an alternative of v, we then have the following formulation

$$\begin{cases} \Delta \left(\sum_{j=1}^{N} \widetilde{\beta}_j \sigma\big(U_j(\boldsymbol{x})\big) \right) = \sum_{i=1}^{K} \hat{\beta}_i \sigma\big(V_i(\boldsymbol{x})\big) \\ \sum_{j=1}^{N} \widetilde{\beta}_j \sigma\big(U_j(\boldsymbol{x})\big) = g(\boldsymbol{x}) \end{cases} \tag{5}$$

with $U_j(\boldsymbol{x}) = \boldsymbol{x}^T \cdot \widetilde{\boldsymbol{W}}^{[j]} + \widetilde{b}_j$ and and N is the number of hidden unit.

For solving the biharmonic equation with the CELM model, a collection of collocation points is sampled from both the interior and boundaries of the domain. At these collocation points, the functions (4) and (5) exhibit coerciveness.

Let Q denote the number of random collocation points in the interior of Ω and denote by $\mathcal{S}_I = \{\boldsymbol{x}_I^q\}_{q=1}^{Q}$, and P denote the number of random collocation points on hyperface of $\partial\Omega$ and denote by $\mathcal{S}_B = \{\boldsymbol{x}_B^p\}_{p=1}^{P}$.

Feeding the collocation points into the pipeline of CELM, we have the following residual terms

$$\begin{cases} \mathcal{R}_v(\boldsymbol{x}_I^q) = \Delta\left(\sum_{i=1}^{K} \widehat{\beta}_i \sigma\big(V_i(\boldsymbol{x}_I^q)\big)\right) - f(\boldsymbol{x}_I^q), \\ \mathcal{R}_u(\boldsymbol{x}_I^q) = \Delta\left(\sum_{j=1}^{N} \widetilde{\beta}_j \sigma\big(U_j(\boldsymbol{x}_I^q)\big)\right) - \sum_{i=1}^{K} \widehat{\beta}_i \sigma\big(V_i(\boldsymbol{x}_I^q)\big), \\ \mathcal{L}_v(\boldsymbol{x}_B^p) = \sum_{i=1}^{K} \widehat{\beta}_i \sigma\big(V_i(\boldsymbol{x}_B^p)\big) - h(\boldsymbol{x}_B^p), \\ \mathcal{L}_u(\boldsymbol{x}_B^p) = \sum_{j=1}^{N} \widetilde{\beta}_j \sigma\big(U_j(\boldsymbol{x}_B^p)\big) - g(\boldsymbol{x}_B^p). \end{cases} \tag{6}$$

To approximate the functions v and u well, we need to adjust the output weights of the CELM model to make each residual of (6) to near-zero or minimize the error of the following system of linear equations:

$$\mathbf{H}\overline{\boldsymbol{\beta}} = \mathbf{S} \tag{7}$$

with

$$\mathbf{H} = \begin{bmatrix} \boldsymbol{O} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & -\mathbf{H}_{22} \\ \mathbf{H}_{31} & \boldsymbol{O} \\ \boldsymbol{O} & \mathbf{H}_{42} \end{bmatrix}, \quad \overline{\boldsymbol{\beta}} = \begin{bmatrix} \widetilde{\boldsymbol{\beta}} \\ \widehat{\boldsymbol{\beta}} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} \boldsymbol{S}_f \\ \boldsymbol{S}_0 \\ \boldsymbol{S}_g \\ \boldsymbol{S}_h \end{bmatrix}.$$

Here,

$$\mathbf{H}_{12} = \begin{bmatrix} \sigma''\left(V_1(\boldsymbol{x}_I^1)\right) & \sigma''\left(V_2(\boldsymbol{x}_I^1)\right) & \cdots & \sigma''\left(V_K(\boldsymbol{x}_I^1)\right) \\ \sigma''\left(V_1(\boldsymbol{x}_I^2)\right) & \sigma''\left(V_2(\boldsymbol{x}_I^2)\right) & \cdots & \sigma''\left(V_K(\boldsymbol{x}_I^2)\right) \\ \vdots & \vdots & \vdots & \vdots \\ \sigma''\left(V_1(\boldsymbol{x}_I^Q)\right) & \sigma''\left(V_2(\boldsymbol{x}_I^Q)\right) & \cdots & \sigma''\left(V_K(\boldsymbol{x}_I^Q)\right) \end{bmatrix} \odot \begin{bmatrix} \sum_{i=1}^{d} \widehat{w}_{i1}^2 & \sum_{i=1}^{d} \widehat{w}_{i2}^2 & \cdots & \sum_{i=1}^{d} \widehat{w}_{iK}^2 \end{bmatrix}$$

$$\mathbf{H}_{21} = \begin{bmatrix} \sigma''\left(U_1(\boldsymbol{x}_I^1)\right) & \sigma''\left(U_2(\boldsymbol{x}_I^1)\right) & \cdots & \sigma''\left(U_N(\boldsymbol{x}_I^1)\right) \\ \sigma''\left(U_1(\boldsymbol{x}_I^2)\right) & \sigma''\left(U_2(\boldsymbol{x}_I^2)\right) & \cdots & \sigma''\left(U_N(\boldsymbol{x}_I^2)\right) \\ \vdots & \vdots & \vdots & \vdots \\ \sigma''\left(U_1(\boldsymbol{x}_I^Q)\right) & \sigma''\left(U_2(\boldsymbol{x}_I^Q)\right) & \cdots & \sigma''\left(U_N(\boldsymbol{x}_I^Q)\right) \end{bmatrix} \odot \begin{bmatrix} \sum_{i=1}^{d} \widetilde{w}_{i1}^2 & \sum_{i=1}^{d} \widetilde{w}_{i2}^2 & \cdots & \sum_{i=1}^{d} \widetilde{w}_{iN}^2 \end{bmatrix}$$

$$\mathbf{H}_{22} = \begin{bmatrix} \sigma\left(V_1(\boldsymbol{x}_I^1)\right) & \sigma\left(V_2(\boldsymbol{x}_I^1)\right) & \cdots & \sigma\left(V_K(\boldsymbol{x}_I^1)\right) \\ \sigma\left(V_1(\boldsymbol{x}_I^2)\right) & \sigma\left(V_2(\boldsymbol{x}_I^2)\right) & \cdots & \sigma\left(V_K(\boldsymbol{x}_I^2)\right) \\ \vdots & \vdots & \vdots & \vdots \\ \sigma\left(V_1(\boldsymbol{x}_I^Q)\right) & \sigma\left(V_2(\boldsymbol{x}_I^Q)\right) & \cdots & \sigma\left(V_K(\boldsymbol{x}_I^Q)\right) \end{bmatrix}$$

$$\mathbf{H}_{31} = \begin{bmatrix} \sigma\left(U_1(\boldsymbol{x}_B^1)\right) & \sigma\left(U_2(\boldsymbol{x}^1)\right) & \cdots & \sigma\left(U_N(\boldsymbol{x}^1)\right) \\ \sigma\left(U_1(\boldsymbol{x}_B^2)\right) & \sigma\left(U_2(\boldsymbol{x}^2)\right) & \cdots & \sigma\left(U_N(\boldsymbol{x}^2)\right) \\ \vdots & \vdots & \vdots & \vdots \\ \sigma\left(U_1(\boldsymbol{x}_B^P)\right) & \sigma\left(U_2(\boldsymbol{x}_B^P)\right) & \cdots & \sigma\left(U_N(\boldsymbol{x}_B^P)\right) \end{bmatrix}$$

and

$$\mathbf{H}_{42} = \begin{bmatrix} \sigma\left(V_1(\boldsymbol{x}_B^1)\right) & \sigma\left(V_2(\boldsymbol{x}^1)\right) & \cdots & \sigma\left(V_K(\boldsymbol{x}^1)\right) \\ \sigma\left(V_1(\boldsymbol{x}_B^2)\right) & \sigma\left(V_2(\boldsymbol{x}^2)\right) & \cdots & \sigma\left(V_K(\boldsymbol{x}^2)\right) \\ \vdots & \vdots & \vdots & \vdots \\ \sigma\left(V_1(\boldsymbol{x}_B^P)\right) & \sigma\left(V_2(\boldsymbol{x}_B^P)\right) & \cdots & \sigma\left(V_K(\boldsymbol{x}_B^P)\right) \end{bmatrix},$$

where σ'' is the second order derivative of activation function σ and $\bigodot$ stands for the Hadamard product operator. In addition, $\mathbf{S}_f = \left(f(\boldsymbol{x}_I^1), f(\boldsymbol{x}_I^2), \cdots, f(\boldsymbol{x}_I^Q)\right)^T$, $\mathbf{S}_0 = \big(\overbrace{0, 0, \cdots, 0}^{Q}\big)^T$, $\mathbf{S}_g = \left(g(\boldsymbol{x}_B^1), g(\boldsymbol{x}_B^2), \cdots, g(\boldsymbol{x}_B^P)\right)^T$ and $\mathbf{S}_h = \left(h(\boldsymbol{x}_B^1), h(\boldsymbol{x}_B^2), \cdots, h(\boldsymbol{x}_B^P)\right)^T$.

When the weights and biases of the hidden layer for the CELM model are randomly initialized and fixed, the matrix $\mathbf{H}$ is determined for given input data, then the output weights $\overline{\beta}$ of CELM model are obtained by solving the linear systems (7). The system equation $\mathbf{H}\overline{\beta} = \mathbf{S}$ is solvable and meets the following several cases:

- If the coefficient matrix $\mathbf{H}$ is square and invertible, then

$$\overline{\boldsymbol{\beta}} = \mathbf{H}^{-1}\mathbf{S}. \tag{8}$$

- If the coefficient matrix $\mathbf{H}$ is rectangular and full rank, then

$$\overline{\boldsymbol{\beta}} = \mathbf{H}^{\dagger}\mathbf{S}. \tag{9}$$

  where $\mathbf{H}^{\dagger}$ is the pseudo inverse of $\mathbf{H}$ defned by $\mathbf{H}^{\dagger} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T$.

- If the coefficient matrix $\mathbf{H}$ is singular, then the Tikhonov regularization[49] can be utilized to to overcome the ill-posedness of (1), and the $\overline{\boldsymbol{\beta}}$ is given by

$$\overline{\boldsymbol{\beta}} = (\mathbf{H}^T\mathbf{H} + \lambda\mathbf{I})^{-1}\mathbf{H}\mathbf{S} \tag{10}$$

  where $\mathbf{I}$ is the identity matrix, and $\lambda > 0$ is the ridge parameter.

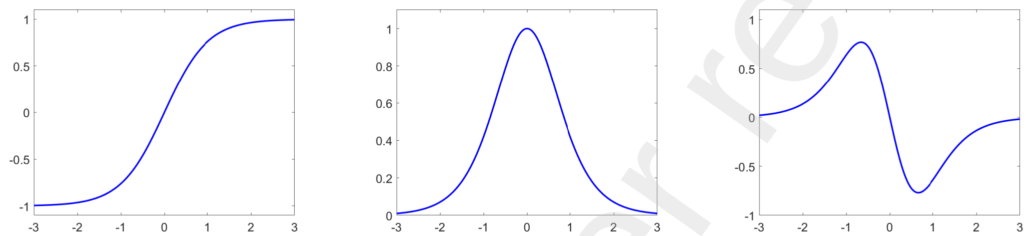
According to the above discussions, the detailed procedure described in the following involved in the CELM algorithm to address the biharmonic equation (1) within finite-dimensional spaces.

Algorithm 1 CELM algorithm for solving biharmonic equation (1).

1) Generating the points set $\mathcal{S}^k$ includes interior points $S_I = \{\boldsymbol{x}_I^q\}_{i=1}^Q$ with $\boldsymbol{x}_I^q \in \mathbb{R}^d$ and boundary points $S_B = \{\boldsymbol{x}_B^p\}_{j=1}^P$ with $\boldsymbol{x}_B^j \in \mathbb{R}^d$. Here, we draw the random points $\boldsymbol{x}_I^q$ and $\boldsymbol{x}_B^p$ from $\mathbb{R}^d$ with positive probability density ν, such as uniform distribution.
2) Selecting a shallow neural network and randomly initialize weights and biases of the hidden layer, then fixing the parameters of the hidden layer.
3) Embedding the input data into the implicit feature space by a linear transformation, then activating nonlinearly the results and casting them into the output layer.
4) Combining linearly the all output features of the hidden layer and output this results.
5) Computing the derivation about the input variable for the output of the CELM model and constructing the linear system(7).
6) Learning the output layer weights by least-squares method.

3.2. The option of activation function and the initialization of weights and biases

Regarding function approximation, the hidden units operated by the activation function in the ELM model can be viewed as a set of basis functions, and the ELM model derives its output through a linear aggregation of these fundamental functions. Similar to other architectures of artificial neural networks (ANN), the selection of a suitable activation function remains crucial for the ELM model. Given that the biharmonic equation (1) considered in this study falls into the category of high-order partial differential equations (PDEs), activation functions with robust regularity are deemed as ideal candidates. In previous works such as Fabiani et al. [50], Calabrò et al. [51], and Ma et al. [52], the authors explored the impact of sigmoidal functions and radial basis functions on the ELM model for solving both linear and nonlinear PDEs. Numerical results indicated that the proposed machine learning method demonstrated commendable numerical accuracy.



(a) Tanh fucntion (b) 1st-order derivative of tanh (c) 2nd-order derivative of tanh

Figure 2: The tanh function and its 1st-order and 2nd-order derivatives, respectively.

Remark 1. (**Lipschitz continuity**) If an activation function σ is continuous (i.e., $\sigma \in C^1$) and satisfies the following boundedness condition:

$$|\sigma(x)| < 1 \quad \text{and} \quad |\sigma'(x)| < 1$$

for any $x \in \mathbb{R}$. Then, we have

$$|\sigma(x) - \sigma(y)| \leqslant |x - y| \quad \text{and} \quad |\sigma'(x) - \sigma'(y)| < |x - y|$$

for any $x, y \in \mathbb{R}$. The activation functions tanh, Gaussian, sine, and the trigonometric function $\sin + \cos$ all fulfill the specified condition and exhibit robust regularity. They ensure the boundedness of the output for basic functions in ELM and enhance the capacity of the ELM model. Figure 2 illustrates the curves of the hyperbolic tangent function and its first-order derivative as well as its second-order derivative. This visualization demonstrates that the derivatives of tanh are both bounded and saturated.

On the other hand, the initialization of weights and biases for the ELM model is crucial. Typically, these weights and biases are initialized based on a uniform distribution. Some studies delve into the examination of the value distribution of internal weights and biases for hidden nodes [53, 54]. Moreover, we extend our investigation to explore the impact of normal distribution initialization on our CELM model and the selection of standard deviation for this initialization approach.

4. Numerical experiments

4.1. Experimental setup

In this section, we apply the proposed CELM algorithm to solve the biharmonic equation (1) in both two and three-dimensional spaces. Two criteria are provided to evaluate the approximation error for our

CELM model with different initialization methods and activation functions. The first criterion is point-wise absolute error

$$APE(\boldsymbol{x}^i) = |\tilde{u}(\boldsymbol{x}^i) - u^*(\boldsymbol{x}^i)|$$

and the second criterion is relative error

$$REL = \sqrt{\frac{\sum_{i=1}^{N'} |\tilde{u}(\boldsymbol{x}^i) - u^*(\boldsymbol{x}^i)|^2}{\sum_{i=1}^{N'} |u^*(\boldsymbol{x}^i)|^2}},$$

where $\tilde{u}(\boldsymbol{x}^i)$ and $u^*(\boldsymbol{x}^i)$ are the approximation obtained by CELM method and exact solution, respectively, for collection set $\{\boldsymbol{x}^i\}_{i=1}^{N'}$, and N' represents the number of collection points. We perform all experimental methods in Numpy (version 1.21.0) on a workstation (256 GB RAM, single NVIDIA GeForce RTX 4090Ti 24-GB).

4.2. Numerical examples

Example 4.1. To start, we aim to approximate the solution to the biharmonic equation (1) within the regular domain $\Omega = [-1, 1] \times [-1, 1]$. An analytical solution and a force term are given by

$$u(x_1, x_2) = \sin(\pi x_1)\sin(\pi x_2)$$

and

$$f(x_1, x_2) = 4\pi^4 \sin(\pi x_1)\sin(\pi x_2),$$

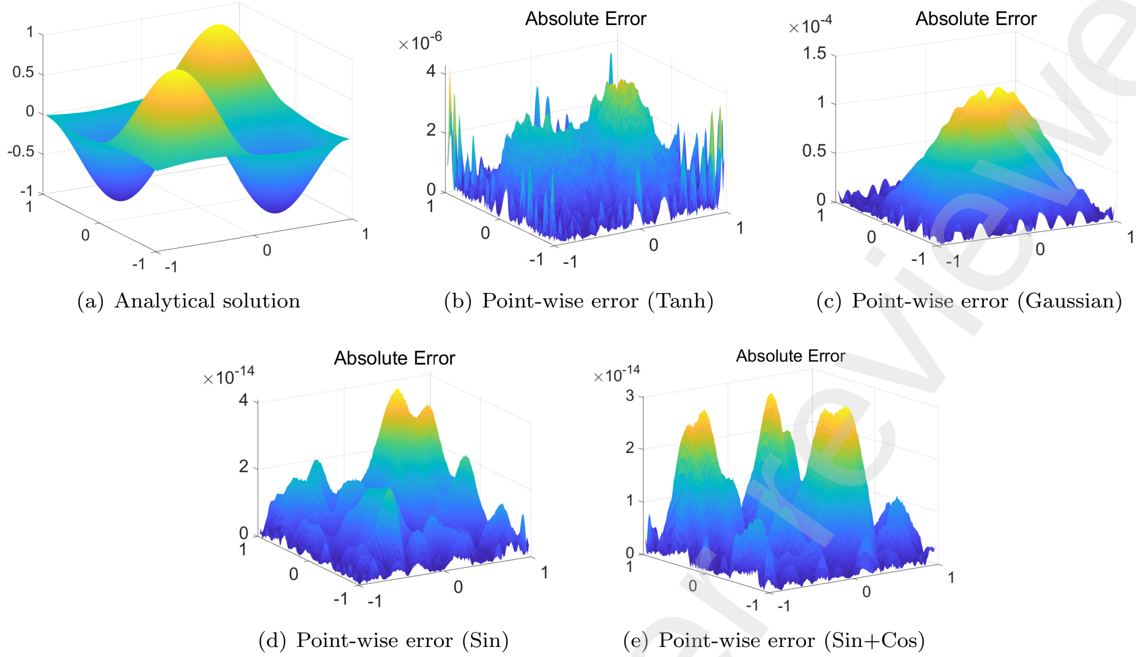
respectively, with the boundary constraints function $g(x_1, x_2) = 0$ and $h(x_1, x_2) = 0$ on $\partial\Omega$.

In this example, we approximate the solution of (1) by the CELM model with 1000 hidden units on 16384 grid points in the domain $[-1, 1] \times [-1, 1]$ with mesh-size $h = 1/128$. Apart from the grid points, the training set includes 10,000 random points sampled from the interior of Ω and 4,000 points randomly sampled from four boundary lines of Ω. Firstly, we perform the proposed CELM algorithm on these collections, the internal parameters(weights and biases) are initialized by a scale uniform distribution initialization in $[-\sigma, \sigma]$ with scale factor $\sigma = 10$. The numerical results are plotted in Figure 3 when activation functions are set as tanh, Gaussian, sin, and trigonometric functions sin+cos, respectively. Secondly, the comparative results of different initialization methods including normal distribution initialization with mean zero and a standard deviation 10, unitary uniform distribution initialization in $[-1, 1]$, scale uniform distribution initialization, and mixed initialization are listed in Table 1. Finally, the curves of the relative error and the running time vary from the hidden nodes, and the curve of the relative error varies from the scale factor σ are depicted in Figure 4 for scale uniform distribution initialization and sine activation function.

Table 1: REL of CELM with various activation functions and initialization of parameters for Example 4.1.

Init Weight	Normal	Unitary uniform	Scale uniform	Normal
Init Bias	Normal	Unitary uniform	Scale uniform	Scale uniform
Tanh	0.0071	1.7823×10^{-9}	3.0943×10^{-6}	1.2773×10^{-4}
Gaussian	0.0621	8.2273×10^{-11}	1.0937×10^{-4}	0.0027
Sin	1.7446×10^{-13}	1.5469×10^{-4}	2.5259×10^{-14}	5.9478×10^{-14}
Sin+Cos	5.8946×10^{-14}	2.7601×10^{-4}	2.0427×10^{-14}	5.2089×10^{-14}

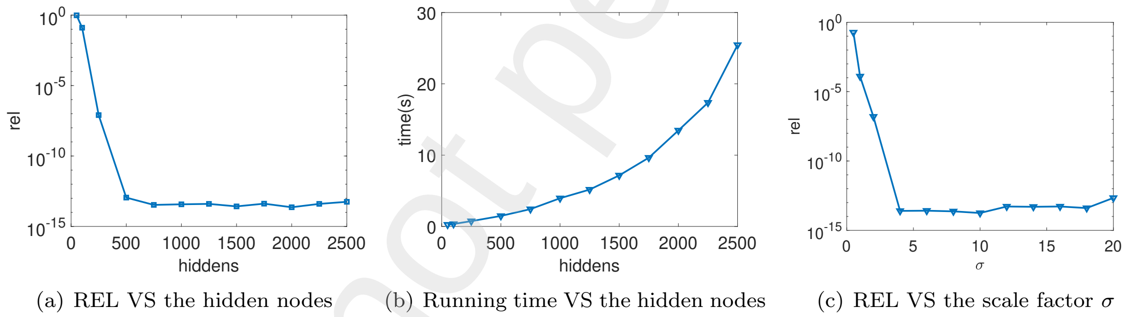
Based on the point-wise absolute error distribution depicted in Figures 3(a) − 3(e), we can infer that CELM models with sine and sine+cosine activation functions outperform those with hyperbolic tangent and Gaussian activation functions. Moreover, the performance of the CELM model with sine activation is competitive with that of the model using sine+cosine activation. The data presented in Table 1 reveals that CELM models with uniformly scaled initialization perform the best for sine and sine+cosine activation functions, surpassing other cases by at least three orders of magnitude in terms of the activation function



(a) Analytical solution (b) Point-wise error (Tanh) (c) Point-wise error (Gaussian)

(d) Point-wise error (Sin) (e) Point-wise error (Sin+Cos)

Figure 3: Numerical results to CELM with scale uniform initialization for Example 4.1.



(a) REL VS the hidden nodes (b) Running time VS the hidden nodes (c) REL VS the scale factor σ

Figure 4: REL VS the hidden nodes (left), running time VS the hidden nodes (middle) and REL VS the scale factor σ (right) to CELM model with scale uniform initialization and sin activation function for Example 4.1.

Table 2: REL and running time of FDM and CELM with sin+cos and scale uniform initialization for Example 4.1.

	number of mesh grid	64×64	128×128	256×256	512×512
FDM	REL	1.55×10^{-3}	3.95×10^{-4}	9.96×10^{-5}	2.51×10^{-5}
	Time(s)	0.224	0.731	31.4	598.39
CELM	REL	4.11×10^{-14}	4.46×10^{-14}	1.21×10^{-14}	1.49×10^{-14}
	Time(s)	3.82	6.72	18.95	80.67

and hidden unit initialization method. As observed in the curves of Figure 4(a), the relative error does not decrease with an increase in the number of hidden units. This implies that the performance of the CELM model with the aforementioned setups tends to stabilize with the growing number of hidden neurons. The time-neuron curve in Figure 4(b) indicates that the runtime exhibits a superlinear increase with the augmentation of hidden neurons, although the rising tendency is gradual. Lastly, Figure 4(c) demonstrates that the CELM model remains stable and robust with variations in the scale factor σ. In summary, our CELM method not only demonstrates high accuracy but also proves to be efficient and robust. Additionally, it's noteworthy that the coefficient matrix in the final linear systems to be solved is full rank, given that the number of samples is greater than that of hidden nodes.

For comparison, we incorporate the finite difference method (FDM) with a five-point central difference scheme to solve the couple formulation (3) of the original problem (1) at various resolutions of mesh grids. The results of both FDM and CELM are presented in Table 2, indicating that our proposed CELM maintains stability across different mesh grid sizes and significantly outperforms FDM in terms of both accuracy and runtime, particularly for dense grids.

Example 4.2. We consider the following biharmonic equation (1) in two-dimensional space and obtain its numerical solution within a hexagram domain Ω, which is derived from the $[0,1] \times [0,1]$. An analytical solution and a force term are given by

$$u(x_1, x_2) = 10(x_1 - 2x_1^3 + x_1^4)(x_2 - 2x_2^3 + x_2^4)$$

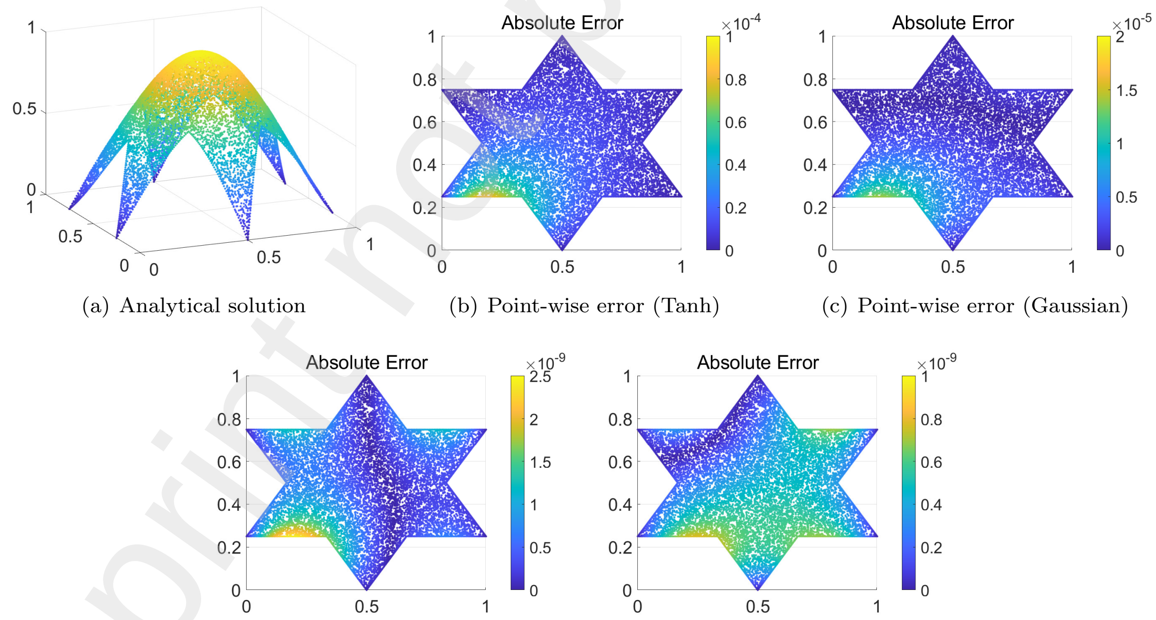
and

$$f(x_1, x_2) = 240(x_1 - 2x_1^3 + x_1^4) + 2880(x_1^2 - x_1)(x_2^2 - x_2) + 240(x_2 - 2x_2^3 + x_2^4),$$

respectively. The boundary functions $g(x_1, x_2)$ can be easily obtained from the exact solution, but for brevity, we will omit it here. By careful calculation, we have the function $h(x_1, x_2)$ on $\partial\Omega$, it is

$$h(x_1, x_2) = 10(12x_1^2 - 12x_1)(x_2 - 2x_2^3 + x_2^4) + 10(x_1 - 2x_1^3 + x_1^4)(12x_2^2 - 12x_2).$$

In this example, the setups of CELM models are the same as the Example 4.1. One can obtain an approximation for the solution of (1) by our proposed CELM algorithm on 20,000 points randomly sampled from the hexagon domain Ω. We plot and list the numerical results in Figures 5 and 6 as well as Table 3.

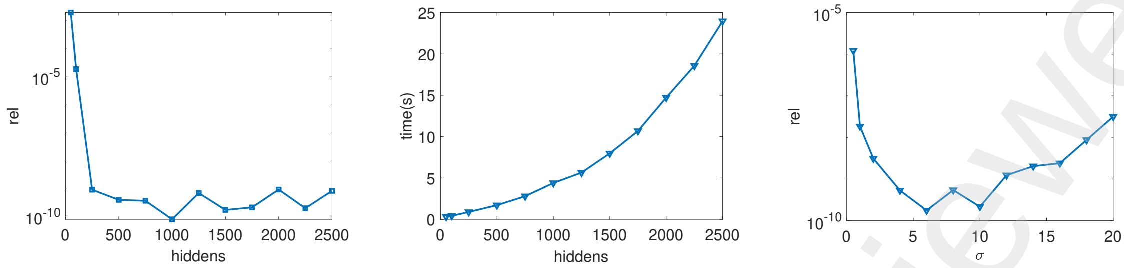


(a) Analytical solution (b) Point-wise error (Tanh) (c) Point-wise error (Gaussian)

(d) Point-wise error (Sin) (e) Point-wise error (Sin+Cos)

Figure 5: Numerical results to CELM with scale uniform initialization for Example 4.2.

Based on the results in Figure 5 and Table 3, the CELM model with sin activation function, on the irregular domain in two-dimensional space, still possesses the capacity to obtain a satisfactory approximation



(a) REL VS the hidden nodes (b) Running time VS the hidden nodes (c) REL VS the scale factor σ

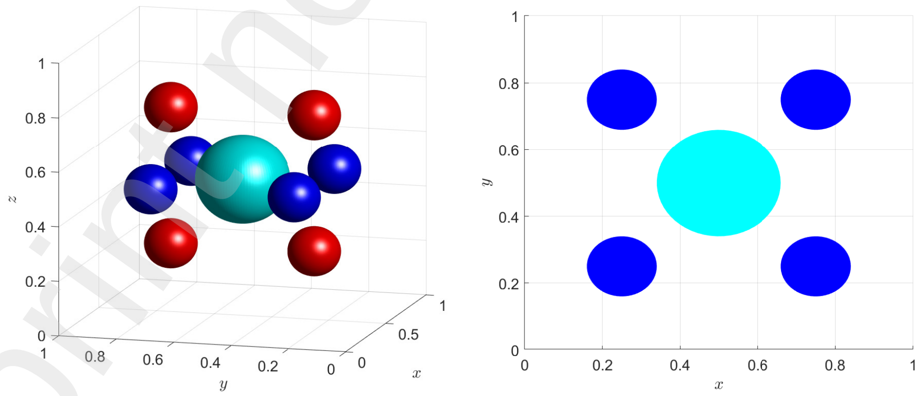
Figure 6: REL VS the hidden nodes (left), running time VS the hidden nodes (middle) and REL VS the scale factor σ (right) to CELM model with scale uniform initialization and sin activation function for Example 4.2.

Table 3: REL of CELM with various activation functions and initialization of parameters for Example 4.2.

Init Weight	Normal	Unitary uniform	Scale uniform	Normal
Init Bias	Normal	Unitary uniform	Scale uniform	Scale uniform
Tanh	7.8642×10^{-4}	9.0761×10^{-9}	3.0579×10^{-5}	0.0011
Gaussian	0.0321	4.6728×10^{-9}	5.0337×10^{-6}	0.0055
Sin	1.9411×10^{-8}	2.2124×10^{-8}	8.9321×10^{-10}	5.0643×10^{-9}
Sin+Cos	5.7731×10^{-9}	4.1012×10^{-9}	6.0818×10^{-10}	7.3112×10^{-9}

for the solution of (1) when the internal parameters of hidden units are initialized by the scale uniform initialization technique, and its performance is superior to that of the ones with other setups. The curves of relative error and running time in Figure 6 demonstrate this CELM model is stable and robust for both the varying number of hidden nodes and the varying scale factor σ. In addition, the coefficient matrix in the final linear systems to be solved is also full rank for this example.

Example 4.3. Let us now approximate the solution of biharmonic equation (1) in a three-dimensional cubic domain $\Omega = [0,1] \times [0,1] \times [0,1]$ with 1 big hole (cyan) and 8 smaller holes (red and blue), see Fig. 7.



(a) Unit cubic domain with some holes (b) An xoy cutting plane of domain at $z = 0.5$.

Figure 7: The interested domain and its cutting plane for Example 4.3.

$$u(x_1, x_2, x_3) = 10(x_1 - 2x_1^3 + x_1^4)(x_2 - 2x_2^3 + x_2^4)(x_3 - 2x_3^3 + x_3^4),$$

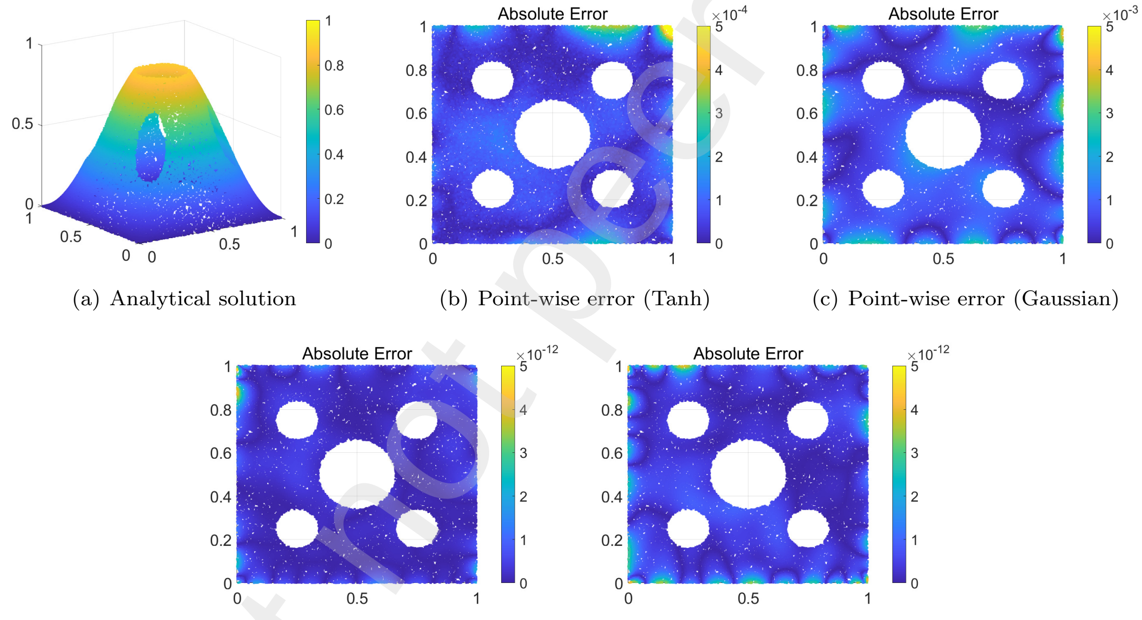
that will naturally lead to the essential boundary function $g(x_1, x_2, x_3)$, and the function of $h(x_1, x_2, x_3)$ is

$$\begin{aligned} h(x_1, x_2, x_3) = {} & 10(12x_1^2 - 12x_1)(x_2 - 2x_2^3 + x_2^4)(x_3 - 2x_3^3 + x_3^4) + 10(x_1 - 2x_1^3 + x_1^4)(12x_2^2 - 12x_2)(x_3 - 2x_3^3 + x_3^4) \\ & + 10(x_1 - 2x_1^3 + x_1^4)(x_2 - 2x_2^3 + x_2^4)(12x_3^2 - 12x_3). \end{aligned}$$

By performing careful calculations, one can determine the expression for the force side, it is

$$\begin{aligned} f(x_1, x_2, x_3) = {} & 240x_2(1 - 2x_2^2 + x_2^3)x_3(1 - 2x_3^2 + x_3^3) + 240x_1(1 - 2x_1^2 + x_1^3)x_3(1 - 2x_3^2 + x_3^3) \\ & + 240x_1(1 - 2x_1^2 + x_1^3)x_2(1 - 2x_2^2 + x_2^3) + 2880x_1(x_1 - 1)x_2(x_2 - 1)x_3(1 - 2x_3^2 + x_3^3) \\ & + 2880x_1(x_1 - 1)x_3(x_3 - 1)x_2(1 - 2x_2^2 + x_2^3) + 2880x_2(x_2 - 1)x_3(x_3 - 1)x_2(1 - 2x_2^2 + x_2^3). \end{aligned}$$

In this example, we obtain the numerical solution of (1) by the CELM model with 2000 hidden units in Ω, the other setups of the CELM model are the same as the Example 4.1. A testing set consists of 80000 random points sampled from $[0, 1] \times [0, 1]$ at $z = 0.5$. Apart from those grid points, the training set also includes 6000 points randomly sampled from six boundaries. We plot and list the numerical results in Figures 8 and 9 as well as Table 4.



(a) Analytical solution (b) Point-wise error (Tanh) (c) Point-wise error (Gaussian)

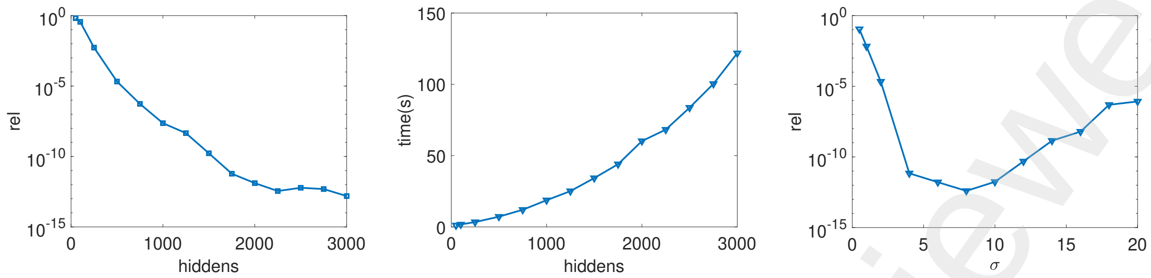
(d) Point-wise error (Sin) (e) Point-wise error (Sin+Cos)

Figure 8: Numerical results to CELM with scale uniform initialization for Example 4.3.

Table 4: REL of CELM with various activation functions and initialization of parameters for Example 4.3.

Init Weight	Normal	Unitary uniform	Scale uniform	Normal
Init Bias	Normal	Unitary uniform	Scale uniform	Scale uniform
Tanh	0.0051	7.1068×10^{-8}	2.7657×10^{-4}	0.0051
Gaussian	0.3977	605164×10^{-9}	0.0047	0.2498
Sin	1.5912×10^{-9}	0.0022	9.4702×10^{-13}	1.3471×10^{-9}
Sin+Cos	1.6214×10^{-9}	0.0015	1.0059×10^{-12}	4.7209×10^{-10}

Based on the point-wise error depicted in Figures 8(b) – 8(e), our proposed CELM model with sine



(a) REL VS the hidden nodes (b) Running time VS the hidden nodes (c) REL VS the scale factor σ

Figure 9: REL VS the hidden nodes (left), running time VS the hidden nodes (middle) and REL VS the scale factor σ (right) to CELM model with scale uniform initialization and sin activation function for Example 4.3.

and sine+cosine activation functions proves to be highly effective in solving the biharmonic equation (1) in three-dimensional space, particularly when the internal parameters of hidden units are initialized using the scale uniform initialization technique. The relative errors presented in Table 4 indicate that the CELM model with the specified configurations outperforms those with other setups. Furthermore, the curves of relative error and running time in Figure 9 showcase the favorable stability and robustness of the CELM model when both the number of hidden nodes and the scale factor σ are increased. It's noteworthy that the coefficient matrix of the final linear systems is also full rank for this example, eliminating the need for adjustment through Tikhonov regularization.

Example 4.4. We consider the following biharmonic equation (1) in there-dimensional space and obtain its numerical solution on a closed domain Ω restrained by two spherical surfaces with radius $r_1 = 0.5$ and $r_2 = 1.0$, respectively. The exact solution is given by

$$u(x_1, x_2, x_3) = \sin(\pi x_1)\sin(\pi x_2)\sin(\pi x_3),$$

that will naturally lead to the determination of the boundary conditions $g(x_1, x_2, x_3)$ and $h(x_1, x_2, x_3)$ on $\partial\Omega$, they are:

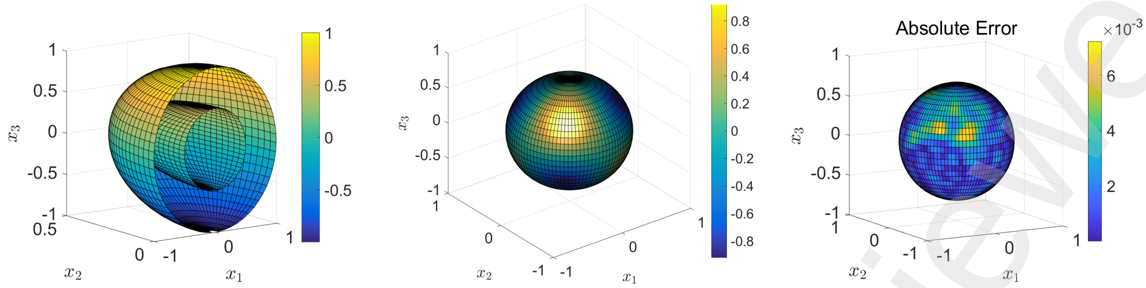
$$\begin{cases} g(x_1, x_2, x_3) = \sin(\pi x_1)\sin(\pi x_2)\sin(\pi x_3), \\ h(x_1, x_2, x_3) = -3\pi^2\sin(\pi x_1)\sin(\pi x_2)\sin(\pi x_3), \end{cases}$$

respectively. By carefully calculations, one can obtain the force side $f(x_1, x_2, x_3) = 9\pi^4\sin(\pi x_1)\sin(\pi x_2)\sin(\pi x_3)$.

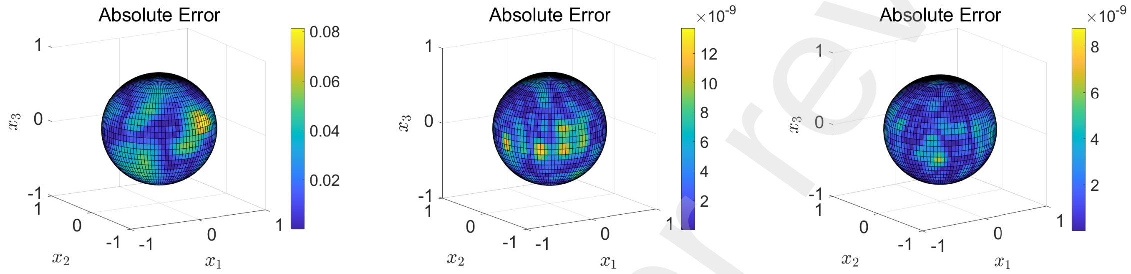
Table 5: REL of CELM with various activation activations and initialization of parameters for Example 4.4.

Init Weight	Normal	Unitary uniform	Scale uniform	Normal
Init Bias	Normal	Unitary uniform	Scale uniform	Scale uniform
Tanh	0.0781	2.5835×10^{-8}	0.0046	0.0391
Gaussian	0.3527	2.1657×10^{-9}	0.0561	0.3491
Sin	1.4873×10^{-5}	0.0037	9.5917×10^{-9}	2.1146×10^{-5}
Sin+Cos	2.0745×10^{-5}	0.0027	5.3452×10^{-9}	1.3402×10^{-5}

In this example, we obtain the numerical solution of (1) by the CELM model with 2000 hidden units in Ω, the other setups of the CELM model are the same as the Example 4.1. The testing set consists of 3000 mesh grid points on the spherical surface with radius $r = 0.8$. Apart from the mesh-grid points, the training set also includes 40000 points sampled randomly from the domain Ω and 20,000 random points sampled equally from the inner and the outer spherical surface, respectively. We plot and list the numerical results in Figures 10 and 11 as well as Table 5.

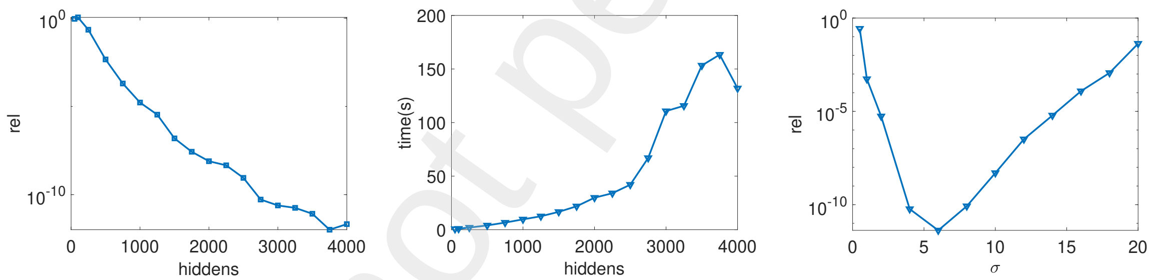


(a) Domain (b) Analytical solution (c) Point-wise error (Tanh)



(d) Point-wise error (Gaussian) (e) Point-wise error (Sin) (f) Point-wise error(Sin+Cos)

Figure 10: Numerical results to CELM with scale uniform initialization for Example 4.4.



(a) REL VS the hidden nodes (b) Running time VS the hidden nodes (c) REL VS the scale factor σ

Figure 11: REL VS the hidden nodes (left), running time VS the hidden nodes (middle) and REL VS the scale factor σ (right) to CELM model with scale uniform initialization and sin activation function for Example 4.4.

For the sphere cavity in three-dimensional space, the point-wise errors in Figures 10(c) – 10(f) reveal that our proposed CELM model with sine and sine+cosine activation functions yields lower errors compared to those with hyperbolic tangent and Gaussian functions in solving the biharmonic equation (1), especially when the internal parameters of hidden units are initialized using the scale uniform initialization technique. The results in Table 5 demonstrate that the CELM model with the specified configurations outperforms those with other setups, as indicated by lower relative errors. Moreover, the curves of relative error and running time in Figure 11 illustrate the favorable stability and robustness of the CELM model when the number of hidden nodes and the scale factor σ are varied. In this example, the coefficient matrix of the final linear systems is also full rank, eliminating the need for adjustment through Tikhonov regularization.

5. Conclusion

We have introduced a coupled extreme learning machine (CELM) to address biharmonic problems by adopting the splitting scheme of the biharmonic equation within a simple ELM network. Similar to other networks, such as PINN and radial basis function models, CELM operates without a mesh and does not rely on an initial guess. In contrast to the PINN structure, CELM proves to be an efficient method for approximating the solution of (1) without necessitating parameter updates through iterative gradient descent-based techniques. Furthermore, we have investigated the influence of activation functions, including tanh, Gaussian, sine, and sine+cosine, on the CELM model, exploring the initialization of internal parameters for hidden units. The computational results provide compelling evidence that the proposed method is highly accurate and efficient in solving (1) in both regular and irregular domains.

In addition, it shall be noted that random projection neural networks have been suggested not only to address the forward problem in steady-state and time-dependent scenarios [55, 56] but have also demonstrated notable efficacy in tackling inverse problems [57, 58]. These networks have exhibited superior performance compared to conventional solvers [55] and alternative machine learning architectures [57], excelling not only in numerical approximation accuracy but also in terms of computational efficiency. Considering the above advantages, we will extend our neural network models to address other PDEs' issues in the future.

Contributions

Xi'an Li: Conceptualization, Methodology, Investigation, Validation, Software and Writing - Original Draft; Jinran Wu: Conceptualization, Methodology, Validation, Writing - review & editing; Zhe Ding: Conceptualization, Investigation, Writing - review & editing; Xin Tai: Writing - Review & Editing, Funding acquisition and Project administration.; Liang Liu: Writing - Review & Editing and Project administration; You-Gan Wang: Writing - Review & Editing and Project administration. All authors have read and agreed to the published version of the manuscript.

Declaration of interests

All authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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